Academic Year: 2026-27

Group No. 97

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

Dhankawadi, Pune -43.

DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING (E&TCE)

A

SYNOPSIS

ON

"AI-Based Semantic Video Analysis and Moment Localization"

Project by

Project Group Members:

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❖ AIM OF PROJECT: -

To develop an intelligent video understanding system capable of identifying, localizing, and retrieving meaningful moments from videos based on semantic user queries while generating concise highlights and contextual information for efficient video analysis.

❖ INTRODUCTION:-

The rapid growth of digital video content from surveillance systems, drones, social media, and other multimedia sources has created a need for efficient video analysis. Conventional video retrieval methods mainly depend on metadata or manual annotations, making it difficult to locate specific moments within long videos. As a result, users often spend considerable time searching for relevant information. Intelligent video understanding techniques aim to overcome these limitations by enabling more accurate and meaningful retrieval of video content based on user intent.

Recent advancements in artificial intelligence and computer vision have made it possible to understand both visual and semantic information from videos. The proposed project focuses on developing a flexible framework that can identify important moments, retrieve relevant video segments, and generate meaningful highlights. The framework is designed to be adaptable, allowing different AI models and techniques to be explored during implementation while supporting applications such as surveillance, disaster management, multimedia search, and smart monitoring.

❖ HARDWARE & SOFTWARE REQUIRMENTS: -

➢ HARDWARE REQUIRED: -

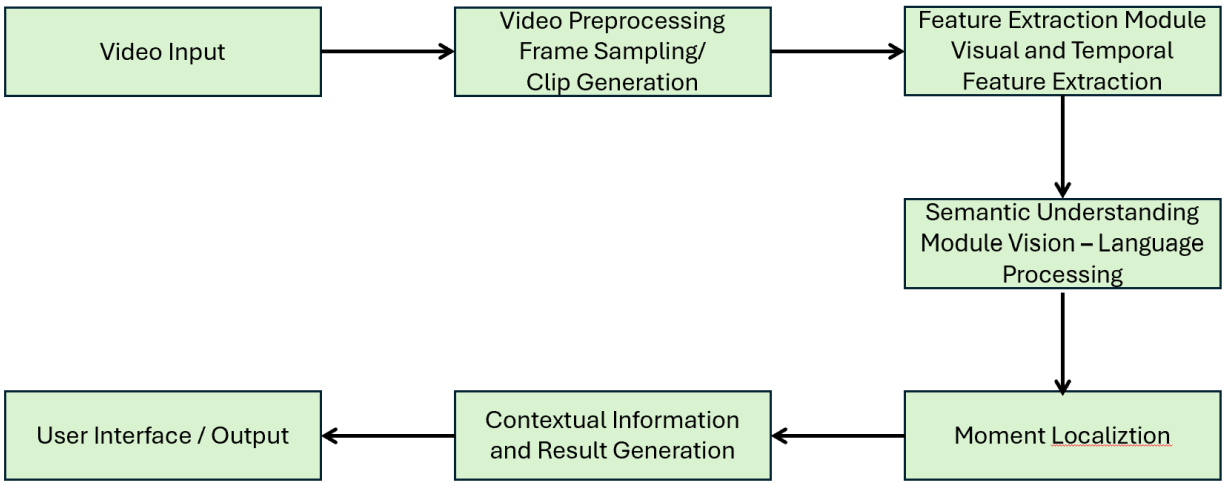
Part	Part Number	Manufacturer
Processor	Intel Core i5 / AMD Ryzen 5	Intel / AMD

RAM	16 GB DDR4	Kingston / Crucial
Storage	512 GB SSD	Samsung / Western Digital
Graphics Card	NVIDIA GeForce RTX Series (CUDA Enabled)	NVIDIA

➢ SOFTWARE REQUIRED: -

1. Python 3.x
2. Pytorch
3. OpenCV
4. FFmpeg
5. CLIP / Equivalent Vision-Language Model
6. Transformer Framework (Hugging Face)
7. Visual Studio Code
8. Jupyter Notebook
9. CUDA Toolkit & cuDNN
10. Git & GitHub

❖ BLOCK DIAGRAM OF PROJECT: -



❖ DESCRIPTION OF BLOCK DIAGRAM :-

- **Video Input**
  - Accepts video data from various sources for analysis.
- **Video pre-processing**
  - This module prepares the input video for analysis. It extracts frames from the video at regular intervals, divides long videos into smaller clips, resizes and normalizes the frames, and removes unnecessary or low-quality data. This ensures that the video is in a consistent format for efficient processing by the AI model.
- **Feature Extraction Module**
  - The feature extraction module identifies important information from the processed video. It captures **visual features** such as objects, people, scenes, and colors, along with **temporal features** like movement, actions, and the sequence of events across frames. These features serve as the foundation for understanding the video content.
- **Semantic Understanding Module**
  - This module analyzes the extracted features to understand the meaning and context of the video. It recognizes activities, interactions, and relationships between objects and people, allowing the system to interpret what is happening rather than simply identifying visual elements. This enables a deeper understanding of the video's content.
- **Moment Localization**
  - The moment localization module identifies the exact time intervals in the video where significant events or actions occur. It predicts the start and end timestamps of relevant moments based on the user's query or the task requirements, making it possible to accurately locate specific events within long videos.
- **Contextual Information & Result Generation**
  - This final module combines the localized moments with contextual information obtained during semantic analysis. It generates meaningful outputs such as event descriptions, summaries, timestamps, highlighted video segments, or alerts. The results are presented in a clear and user-

friendly format, allowing users to quickly access the most relevant information from the video.

- **User Interface / Output**
  - Displays the retrieved moments, highlights, and final output to the user.

❖ STEP- WISE EXECUTION: -

STEP 1 (July)	▪ Finalization of project domain and project topic. ▪ Literature survey and study of related research papers. ▪ Design of project architecture and block diagram. ▪ Preparation of synopsis.
STEP 2 (August)	▪ Study of video understanding, computer vision, and multimodal AI concepts. ▪ Selection of software tools, datasets, and development environment.
STEP 3 (September)	▪ Collection and preprocessing of video datasets. ▪ Design of the system workflow. ▪ Implementation of video preprocessing and feature extraction modules.
STEP 4 (October)	▪ Development of semantic understanding and moment localization modules. ▪ Preparation and submission of seminar report.
STEP 5 (December)	▪ Implementation of highlight generation and result generation modules. ▪ Integration of all system components. ▪ Initial testing of the proposed system.
STEP 6 (Jan-Feb)	▪ Performance evaluation and optimization of the system. ▪ Refinement of the user interface and output generation.
STEP 7 (Feb-March)	▪ System testing using different video datasets. ▪ Second assessment submission.
STEP 8 (April)	▪ Preparation and writing of the final project report. ▪ Verification, formatting, and printing of the report.
STEP 9 (May)	▪ Final project demonstration and presentation. ▪ Submission of the final project report.

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